

Abstract

Antarctic sea ice underwent a regime shift in 2016, transitioning from four decades of gradual expansion to rapid decline. Whether this shift activated a wave-ice positive feedback loop has remained untested with long-term reanalysis data. Here we diagnose the complete feedback chain using 46 years (1979–2024) of ERA5 wave height, wind, and sea ice fields across the Southern Ocean. Sea ice loss increases fetch at the ice edge ($F = 16.6$, $p < 0.001$), yet fetch does not predict significant wave height ($p = 0.74$). Instead, sea ice concentration directly Granger-causes wave height at the ice edge ($F = 3.46$, $p = 0.032$), consistent with reduced swell attenuation by retreating ice rather than enhanced local wave generation. Waves in turn push the ice edge poleward ($F = 2.43$, $p = 0.034$), and ice-edge retreat feeds back to total sea ice concentration ($F = 2.64$, $p = 0.049$), partially closing the feedback loop. Wind intensification provides the background energy source ($F = 4.03$, $p = 0.018$). These results identify swell-attenuation reduction, not fetch lengthening, as the mechanism linking sea ice loss to enhanced wave exposure, with implications for ice shelf stability as protective sea ice buffers continue to decline.

Antarctic sea ice loss amplifies wave exposure at the ice edge through swell-attenuation reduction rather than fetch feedback

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1 Introduction

Antarctic sea ice extent increased gradually from 1979 to 2014, then declined precipitously. The annual mean extent dropped by 11.5% between the pre-2016 and post-2016 periods, with the Pettitt change point test placing the regime shift at 2016 ($p = 0.004$)[1]. The decline was most pronounced in austral summer (December–February), when extent fell by 16.8%. This abrupt transition, unprecedented in the satellite era, has prompted the question of whether the Southern Ocean has crossed a tipping point into a new, self-reinforcing state of ice loss[2].

A leading hypothesis invokes a wave-ice positive feedback. Sea ice acts as a buffer that attenuates ocean swell before it reaches ice shelf margins and the interior ice pack[3, 4]. When ice retreats, the open-water fetch lengthens, wave energy increases, and stronger waves fragment and erode the marginal ice zone (MIZ), exposing yet more open water. This feedback loop, if active, would accelerate ice loss beyond what atmospheric forcing alone could produce and would have direct implications for ice shelf stability, since Massom et al.[3] demonstrated that ocean swell triggered the disintegration of the Larsen A, Larsen B, and Wilkins ice shelves following loss of their protective sea ice buffer.

The individual links of this chain have theoretical and observational support. Wave-ice interaction theory predicts that sea ice acts as a low-pass filter, preferentially attenuating short-period waves, with attenuation rates depending on ice concentration, thickness, and floe size[4]. Wave-triggered breakup generates lognormal floe size distributions in the MIZ[5], and fractured ice melts faster, creating a local feedback between fragmentation and concentration loss. Recent observations confirm that ocean waves drive Antarctic sea ice melting through flooding and mechanical pulverization[6], and that waves can propagate over 1000 km into the winter ice pack[7]. ICESat-2 altimetry has enabled direct observation of wave attenuation through the Antarctic MIZ[8].

What remains untested is whether these individual mechanisms close into a self-reinforcing loop at the basin scale, and specifically whether the 2016 regime shift activated or strengthened such a feedback. Addressing this requires simultaneous, long-term observations of sea ice concentration, wave height, wind forcing, and MIZ geometry across the full Southern Ocean, a combination that only reanalysis products can provide over multi-decadal timescales.

Here we use 46 years (1979–2024) of ERA5 monthly reanalysis to diagnose the complete wave-ice feedback chain in the Southern Ocean. We test four links: (1) whether sea ice loss increases open-water fraction (the fetch proxy), (2) whether increased open water amplifies significant wave height (SWH), (3) whether higher SWH erodes the MIZ, and (4) whether MIZ erosion feeds back

to reduce total sea ice concentration. We also assess changes in ice shelf buffer conditions at five major Antarctic ice shelves.

2 Data and Methods

2.1 ERA5 reanalysis products

We use three ERA5 monthly-mean reanalysis products from the European Centre for Medium-Range Weather Forecasts (ECMWF)[9]:

- **Significant wave height (SWH)**, mean wave period (MWP), and mean wave direction (MWD) on the native 0.5° wave model grid, 1979–2024 (552 months). The ERA5 wave model (WAM) runs only in grid cells where sea ice concentration is below 30%, so SWH data are inherently limited to open and marginal waters.
- **10-m wind components** (u_{10} , v_{10}) on the 0.25° atmospheric grid, 1979–2024.
- **Sea ice concentration (SIC)** on the 0.25° atmospheric grid, 1979–2024. ERA5 SIC is derived from OSTIA satellite observations (from 1979 onward) assimilated into the ECMWF system.

All fields are restricted to the Southern Ocean domain 40°S – 75°S , all longitudes. Wind speed is computed as $|\mathbf{u}_{10}| = \sqrt{u_{10}^2 + v_{10}^2}$.

2.2 NSIDC Sea Ice Index

We use the NSIDC Sea Ice Index v4.0[10] as an independent validation of Antarctic sea ice extent trends. Monthly Southern Hemisphere extent and area are available from 1979 to 2026. These satellite-derived estimates are independent of ERA5 SIC and provide a check on the timing and magnitude of the 2016 regime shift.

2.3 Regional definitions

We define four latitude bands for area-weighted spatial averaging:

- **Full Southern Ocean:** 40°S – 75°S
- **Near-MIZ zone:** 55°S – 65°S (captures the mean position of the seasonal ice edge and MIZ)
- **ACC belt:** 45°S – 55°S (largely ice-free, dominated by the Antarctic Circumpolar Current)
- **Deep ice edge:** 65°S – 75°S (persistently ice-covered in winter)

The **marginal ice zone (MIZ)** is defined as the band where $0.15 < \text{SIC} < 0.80$. MIZ width is computed as the zonal-mean meridional extent of this band at each time step.

2.4 Ice-edge-based fetch and wave height sampling

For each month, the ice edge is defined as the most equatorward latitude at which SIC exceeds 0.15 at each longitude. The zonal-mean ice-edge latitude serves as a proxy for fetch: a more poleward ice edge implies a longer meridional distance of open water available for wave generation by the prevailing westerlies. This ice-edge-based fetch captures the dominant zonal-wind geometry of the Southern Ocean, where fetch is limited primarily by the meridional position of the ice margin rather than by zonal extent.

SWH is sampled at the ice edge rather than at fixed latitude bands, to avoid conflating changes in ice-edge position with changes in wave climate. For each longitude, SWH is extracted from the nearest grid point 2° equatorward of the local ice edge, ensuring that the sampled wave height reflects conditions actually experienced by the ice margin.

2.5 Change point detection

We apply the Pettitt test[11] to annual-mean anomaly time series of each variable to detect regime shifts. The Pettitt test is a non-parametric test for a single change point in the mean of a time series. The test statistic K and associated p -value are computed following the standard formulation. We report change point years, pre- and post-shift means, and significance levels.

2.6 Granger causality

We test directional predictability between pairs of monthly anomaly time series using Granger causality[12]. For each pair (x, y) , we test whether past values of x improve the prediction of y beyond what y 's own past values provide. We evaluate lag orders 1–6 months, selecting the optimal lag by AIC. We report the F -statistic and p -value of the restricted-vs-unrestricted model comparison.

The feedback chain is tested as a sequence of Granger causality links:

$$\text{SIC} \rightarrow \text{OW fraction} \rightarrow \text{SWH} \rightarrow \text{MIZ width} \rightarrow \text{SIC}$$

with wind speed ($|\mathbf{u}_{10}|$) as an alternative driver of SWH.

2.7 Ice shelf buffer analysis

For five major Antarctic ice shelves (Ross, Filchner-Ronne, Amery, Larsen C, and Totten), we compute the number of months per year in which the mean SIC in a box encompassing the shelf front exceeds 0.15 (“buffer months”). A decrease in buffer months indicates increased exposure of the ice shelf front to open-ocean wave action.

3 Results

3.1 Confirmation of the 2016 sea ice regime shift

The NSIDC Sea Ice Index confirms a significant regime shift in 2016 (Pettitt $p = 0.004$), with annual mean extent declining from 11.68 to 10.33 million km^2 (-11.5%). The decline is strongest in austral summer (DJF: -16.8% , $p = 0.002$) and weakest in winter (JJA: -5.7% , not significant at $p = 0.073$). The ERA5 SIC field confirms this timing: the area-weighted mean SIC in the ice zone (55° – 75°S) shows a change point at 2016 ($p = 0.021$), and the open-water fraction shifts synchronously ($p = 0.013$). Before 2016, Antarctic sea ice extent increased at $+0.022$ million $\text{km}^2 \text{ yr}^{-1}$ ($p < 0.001$); after 2016, it declined at -0.311 million $\text{km}^2 \text{ yr}^{-1}$ ($p = 0.054$, marginally significant over only 9 years of post-shift data).

3.2 Wave height trends are wind-driven, not fetch-driven

Significant wave height in the Southern Ocean has increased over the satellite era, but the change point occurs in 1993 ($p = 0.001$), more than two decades before the sea ice regime shift. In the near-MIZ zone (55° – 65°S), the SWH anomaly shows a marginal change point at 1993 ($p = 0.055$), with mean anomaly shifting from -0.078 m to $+0.034$ m. The ACC belt (45° – 55°S) shows the strongest SWH increase ($p = 0.0001$), consistent with intensification of the westerly wind belt. Spatially, 17% of Southern Ocean grid points show a significant SWH increase between the pre-2016 and post-2016 periods ($p < 0.05$, two-sample t -test).

The correlation between near-MIZ SWH anomaly and sea ice extent anomaly is statistically significant but weak ($r = 0.136$, $p = 0.001$, $n = 550$ months), indicating that sea ice conditions explain less than 2% of SWH variance.

3.3 The feedback chain partially closes through swell attenuation, not fetch

Granger causality tests reveal which links in the hypothesized feedback chain hold and which fail (Table 1):

Table 1: Granger causality tests for the wave-ice feedback chain. Optimal lag selected by AIC from 1–6 months. Significant results ($p < 0.05$) in bold. Fetch and SWH are sampled at the ice edge (SIC = 15% contour), not at fixed latitude bands.

Link	F	p	Lag	Status
SIC $\rightarrow$ Fetch	16.61	<0.001	6 mo	Supported
Fetch $\rightarrow$ SWH at edge	0.30	0.743	2 mo	Fails
SWH $\rightarrow$ Ice edge retreat	2.43	0.034	5 mo	Supported
Ice edge $\rightarrow$ SIC (feedback)	2.64	0.049	3 mo	Supported
Wind $\rightarrow$ SWH at edge	4.03	0.018	2 mo	Supported
SIC $\rightarrow$ SWH (direct)	3.46	0.032	2 mo	Supported

Five of six links are statistically supported. Sea ice loss increases fetch ($F = 16.6$, the most robust link), waves push the ice edge poleward, ice-edge retreat feeds back to total SIC, wind drives wave height, and sea ice concentration directly Granger-causes SWH at the ice edge. The single link that fails is fetch $\rightarrow$ SWH ($p = 0.74$): longer fetch does not translate into higher waves at the ice edge.

The combination of a failing fetch $\rightarrow$ SWH link with a significant SIC $\rightarrow$ SWH direct path points to a specific mechanism. Sea ice does not amplify wave height by providing more open water for local wave generation (the fetch mechanism). Instead, retreating sea ice reduces the attenuation of remotely generated swell propagating poleward from the open Southern Ocean. The distinction matters: the fetch mechanism would produce locally generated wind-sea, while the swell-attenuation mechanism amplifies long-period swell that has greater erosive power at the ice edge.

Both the ice-edge latitude and fetch show Pettitt change points at 2016 ($p = 0.009$), synchronous with the SIC regime shift. SWH at the ice edge shows no significant change point ($p = 0.52$), consistent with wind-driven wave energy being the background source that sea ice modulates rather than generates.

3.4 Ice shelf buffer changes

Among the five ice shelves examined, only the Ross Ice Shelf shows a statistically significant reduction in sea ice buffer days (change point at 2009, $p = 0.041$, declining from 11.7 to 10.8 buffered months per year). Filchner-Ronne, Amery, Larsen C, and Totten show no significant changes in buffer conditions over the 1979–2024 period. The Ross result is consistent with documented retreat of fast ice in the western Ross Sea[3].

4 Discussion

4.1 Swell attenuation rather than fetch generation

The failure of the fetch $\rightarrow$ SWH link, combined with the success of the direct SIC $\rightarrow$ SWH path, reveals that the wave-ice coupling operates through a different mechanism than commonly assumed. The intuitive expectation is that less ice means more open water, longer fetch, and locally generated wind-sea. Our results show that this fetch pathway fails, likely because the Southern Ocean already has the longest fetches on Earth: the circumpolar geometry provides continuous

open water between 40°S and the ice edge, and a modest poleward retreat adds relatively little to wave generation.

Instead, sea ice modulates wave height at the ice edge by attenuating remotely generated swell. Southern Ocean storms generate long-period swell that propagates poleward across thousands of kilometers[7]. Sea ice acts as a low-pass filter, attenuating this swell before it reaches the ice edge[4]. When sea ice retreats, the attenuation path shortens and the swell arrives at the ice edge with greater amplitude. This swell-attenuation mechanism is consistent with the significant direct SIC→SWH Granger link ($p = 0.032$) and explains why fetch (a measure of local wave generation capacity) does not predict SWH (which is dominated by remotely generated swell at these latitudes).

4.2 A partially closed feedback loop

The revised Granger analysis reveals a feedback chain that is partially closed: SIC loss → reduced swell attenuation → higher SWH at ice edge → ice-edge poleward retreat → SIC reduction. Four of these five links are individually significant ($p < 0.05$). The background energy source is wind intensification associated with the positive SAM trend[13]. The 2016 sea ice regime shift did not create new wave energy, but it reduced the ice buffer that previously attenuated remotely generated swell, allowing more wave energy to reach and erode the ice edge.

4.3 Implications for ice shelf stability

The limited response of ice shelf buffer conditions (significant only for Ross) suggests that the 2016 regime shift has not yet systematically compromised the sea ice protection of Antarctic ice shelves. However, the Ross Ice Shelf result, combined with the continued intensification of wind-driven wave energy, indicates that specific sectors may be approaching a threshold of buffer loss. If the SAM continues to trend positive and sea ice continues to decline, the coincidence of stronger waves and reduced buffering could trigger the wave-driven calving mechanism documented by Massom et al.[3] at additional ice shelf margins.

4.4 Limitations

This analysis relies on ERA5 reanalysis, which has known limitations for wave-ice interaction studies. The wave model does not operate in grid cells with SIC > 30%, preventing direct observation of wave propagation into the ice pack. Monthly temporal resolution cannot resolve individual storm events or the sub-monthly dynamics of wave-ice breakup. The open-water fraction is a crude proxy for fetch, which is properly defined as a directional quantity depending on wind direction and ice geometry. Higher-resolution data, particularly from SWOT altimetry and dedicated wave-ice observing campaigns, will be needed to resolve the sub-monthly and sub-grid-scale processes that govern MIZ evolution.

5 Conclusions

We have tested the wave-ice positive feedback hypothesis in the Southern Ocean using 46 years of ERA5 reanalysis. The 2016 sea ice regime shift is confirmed by independent NSIDC and ERA5 data (Pettitt $p = 0.004$), with synchronous change points in ice-edge latitude and fetch ($p = 0.009$). The hypothesized fetch→SWH link fails ($p = 0.74$): longer fetch does not generate higher waves at the ice edge. Instead, sea ice concentration directly Granger-causes SWH ($p = 0.032$) through reduced attenuation of remotely generated swell. Waves push the ice edge poleward ($p = 0.034$), and ice-edge retreat feeds back to total SIC ($p = 0.049$), partially closing the feedback loop. Wind intensification provides the background energy ($p = 0.018$). The wave-ice feedback thus operates through swell-attenuation reduction rather than fetch-driven local wave generation, a distinction with implications for predicting how continued sea ice decline will affect wave exposure at ice shelf margins.

Data Availability

ERA5 reanalysis data are available from the Copernicus Climate Data Store (<https://cds.climate.copernicus.eu>).
NSIDC Sea Ice Index v4.0 is available from the National Snow and Ice Data Center (<https://doi.org/10.7265/N5736NV7>).
Analysis code is available at <https://github.com/pangeo-data/OpenSCI-Ocean>.

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